S3 Supplementary QC and Robustness

Support, missingness, matching collapse, and reviewer-facing robustness summary

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Short title: S3 QC - Apollo propofol occupancy

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S3.0 Sample and file coverage

20 participants, 80 recordings, four protocol phases each. G0 inventory and G1 join: 80/80 files matched to `datainfo.mat`. G2 H1/H2 support: 80/80 recordings met ≥ 30 valid timepoints and $\geq 50\%$ valid. Occupancy eligible-window counts are in `results/v1_occupancy_20260921/occupancy_subject.csv` (`n_eligible_*` columns; typically 60–105 windows per recording).

S3.1 Identity and scaling QC

Filename identity is `sub-NNN_sedation_rest_run-RRR_acq-AAA`. Join keys are original Cambridge filename keys (subject + acq/run), not FieldTrip `run-1..4`. Scaling of the occupancy frame uses all baseline rows finite in all eight features, with a within-participant baseline median/MAD; one-step eligibility is applied separately. Non-finite or $\leq 10^{-8}$ scales become invalid (`NaN`); no MAD floor is clamped in. ComBat is not applied. Exported `features_robust_z` medians are 0 in every recording — a positive control that local robust-z cannot carry occupancy.

S3.2 Matching and testability

Baseline-anchor matching: 7/20 participants with ≥ 20 pairs. The $n = 10$ subject gate is not lowered. Local-chart matching included 19/20 but is not a cross-phase physical match (S2.4). L2/L3 complete-case n at first sedation is 4 (NOT_TESTABLE). LOR group contrasts: impaired arm $n = 7$ (NOT_TESTABLE). ROR occupancy contrast: not-returned $n = 0$.

S3.3 Bootstrap and sign-test QC

Occupancy bootstrap uses 10 000 participant resamples, seed 20260920. First-sedation reporting contrast median Δ CI $[0.56, 2.40]$; d_z CI $[0.82, 1.69]$. These intervals and the sign tests are descriptive/exploratory; no confirmatory primary endpoint was locked. Sign-test zeros: none in the three occupancy contrasts. Two first-sedation decreases (sub-014, sub-028) remain in the table and in the sign denominator.

S3.4 Label QC

Occupancy-blind LOR labels recover 7/13, matching Chennu et al. Examples that protocol 3 is not automatically an LOR label: sub-002 later $k = 39/40$ (responsive); sub-003 later $k = 3/40$ (impaired). Package `unresponsive` would have labelled both.

S3.5 Reviewer-facing robustness summary

Robustness summary

The occupancy trajectory is within-subject, uses a baseline-only frame, and is consistent in 18/20 participants at first sedation and 20/20 at later sedation. The matched-state question fails a pre-specified support gate rather than being rescued by lowering n. The exploratory LOR association analysis uses occupancy-blind labels that recover the published 7/13 split but does not establish an absence of association; the impaired arm has $n = 7$. The isolated robustness branch retained 20/20 paired participants for all 11 tested variants and found first-sedation d_z values from 1.049 to 1.228, with no variant-specific support failure. Feature shares and scale alternatives are computational diagnostics, not biological validation. Remaining limits are the exploratory status (G3/G4 remain open), post-hoc occupancy promotion, and the fact that untested preprocessing, feature sets, and cohorts may behave differently.

Bibliography